

Abel Zelalem

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Professional Summary:

Dedicated Computer Engineering student with hands-on experience in embedded systems, robotics programming, and hardware/software co-design. Skilled in C++, SystemVerilog, and digital logic, with applied experience in industrial robot programming and microcontroller-based systems. Recognized for strong problem-solving, adaptability, and the ability to contribute effectively in both independent and team-based engineering environments. Experienced in applying engineering concepts to real-world automation.

Education:

University of Nevada, Las Vegas

Expected Graduation: May 2027

Bachelor of Science: Computer Engineering

College of Southern Nevada

May 2024

Associate of Science

Technical Skills

Languages: C | C++ | SystemVerilog | Verilog | RAPID | Arduino | Java (Familiar)

Architectures: RISC-V | ARM | AVR | MIPS (Familiar)

Hardware and Circuits: Microcontrollers (ATmega328P) | Digital Logic Design | Circuit Analysis | Analog Circuits

Applications: Git | Microchip Studio | ModelSim | Quartus II | LTspice | Visual Studio Code | MATLAB | Replit

Technologies: File handling, Sorting Algorithms, Data analysis

Project Experience:

Digital Logic Design II, SystemVerilog Elevator

April 2025

- Engineered and simulated a multi-floor elevator control system from the ground up using SystemVerilog.
- Implemented logic for floor requests, directional movement, and door operations within a structured state machine framework.
- Conducted rigorous verification and timing analysis in ModelSim, ensuring functional accuracy under diverse test scenarios in digital logic design.

Circuits II, Pulse Width Modulator (PWM) Circuit

August 2025

- Designed and built a PWM circuit using a 555 timer and supporting analog components to achieve variable duty cycle control.
- Performed oscilloscope-based measurements to validate frequency and duty cycle performance, aligning results with theoretical predictions.
- Strengthened skills in analog circuit design, LTspice simulation, and real-world signal analysis through hands-on experimentation.

Relevant Experience:

Robotics Programmer

February 2026 - Present

Naama Hair and Wigs LLC

Las Vegas, NV

- Programmed and operated the ABB YuMi IRB 14000 using RAPID (with initial development in RobotStudio) to perform precise pick-and-place tasks, including coordinate system calibration, maintenance, and troubleshooting.
- Designed and prototyped custom 3D fixtures and tooling to enable reliable robotic handling and alignment, directly supporting automation in the wig manufacturing process.

Store Associate

November 2021 - November 2023

In-N-Out Burger

Las Vegas, NV

- Collaborated closely with In-N-Out team members, fostering a cohesive work environment centered on effective communication and mutual support.
- Demonstrated efficiency in time management, balancing responsibilities during peak hours to ensure smooth operation and customer satisfaction.